

GrafoPropagation v29: A Self-Supervised Geometric Transformer for Molecular Property Prediction with Spherical Shock Terrain and Borel–Laplace Resurgent Optimization

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Abstract

We present GrafoPropagation v29, a novel geometric transformer architecture for the regression of the HOMO–LUMO gap on the QM9 dataset. The model departs from conventional E(3)/S(3)-symmetric networks and instead relies on three core innovations: (i) an asymmetric self-supervised pre-training phase based on DeepMind’s BYOL, which prevents dimensional collapse; (ii) a **Spherical Shock Terrain** module that encodes atomic positions through a softmax-based topological competition and a shared global sphere state; and (iii) a **Borel–Laplace Resurgent Optimizer**, which replaces gradient clipping with an analytic mapping of divergent gradients onto the Borel plane. Trained exclusively on the standard QM9 split (110k training molecules), the model achieves a **test MAE of 0.0576 eV**, **RMSE of 0.0824 eV**, **R² of 0.99594**, and a **Pearson correlation of 0.99801**, using only 2.84 million parameters. The complete source code, pre-trained checkpoints, and execution instructions are publicly available, requiring only two dependency installations to run on any Kaggle environment.

1 Introduction

The prediction of quantum chemical properties from molecular structure has been dominated by graph neural networks (GNNs) and transformers that explicitly enforce rotational and permutational symmetries. While physically motivated, such constraints increase architectural complexity and can hinder the learning of emergent representations.

In this work, we take a radically different approach. The architectural logic was first sketched by the author—a non-programmer with a sixth-grade formal education—as a conceptual drawing in a simple paint program. This drawing was then handed to large language models (DeepSeek, Gemini, ChatGPT), which, through iterative dialogue, translated the visual logic into a mathematically rigorous and fully executable codebase. The result is **GrafoPropagation v29**, a model that achieves state-of-the-art regression accuracy on the QM9 HOMO–LUMO gap without any explicit equivariance constraints.

This paper is the chemical counterpart of a previous work by the same author in Natural Language Processing: “*GrafoPropagation NLP: A Graph-Based Reasoning Architecture*” [1]. The unifying philosophy is the treatment of input tokens (words or atoms) as nodes in a fully connected graph, over which geometric attention, accelerated Rotary Position Embeddings (RoPE), and XOR-based fusion extract high-quality representations.

2 Model Architecture

The training pipeline consists of three sequential phases, all restricted to the training partition to guarantee zero data leakage.

2.1 Asymmetric BYOL Pre-Pre-Training

We adopt DeepMind’s Bootstrap Your Own Latent (BYOL) [2] as the self-supervised pre-pre-training objective. Two networks—a Student and a Teacher—process differently augmented views of the same molecule. The Student learns to predict the Teacher’s latent representation, while the Teacher is updated via Exponential Moving Average (EMA) without back-propagation. This asymmetric gradient flow prevents representational collapse without requiring negative samples.

The augmentation consists of random rotations of the Spherical Shock Terrain (Section 2.2), with a curriculum that progressively increases the rotation amplitude from 0.15π to π over 12 epochs.

2.2 Spherical Shock Terrain

Inspired by the principles “*what I feel, everyone feels*” and “*you may fit, but you may not be fitted*”, this module projects atomic coordinates onto a sphere of reference points (generated by the Fibonacci sphere algorithm) and produces an intensity map that is added to the initial token embeddings.

Let $\mathbf{p}_i \in \mathbb{R}^3$ be the position of atom i , and let $\mathbf{a}_j \in \mathbb{R}^3$ be the j -th anchor point on the sphere. The raw shock intensity is computed as

$$s_{ij} = g_j \exp \left(-\frac{\|\mathbf{p}_i - \mathbf{a}_j\|^2}{2\sigma_j^2} \right), \quad (1)$$

where g_j and σ_j are learnable gain and bandwidth parameters. Two mechanisms then operate:

1. **Topological Competition:** A softmax along the anchor dimension enforces that atoms compete to associate with distinct terrain points.
2. **Global Sphere State:** The per-atom fit probabilities are aggregated across the molecule, normalized, and broadcast back, providing a collective geometric context.

The final shock tensor is added to the token embeddings via a learnable linear projection.

2.3 Borel–Laplace Resurgent Optimizer

Standard gradient clipping truncates the loss landscape, potentially discarding valuable curvature information. We propose a **Borel–Laplace Resurgent Optimizer**, a wrapper around AdamW that processes the gradient before each update step.

Given a global gradient norm G , the raw gradient is scaled by

$$\alpha(G) = \frac{1}{1 + \eta G} + e^{-1/(\eta G + \epsilon)} \cdot \frac{1}{G + \epsilon}, \quad (2)$$

where η is the current learning rate. For large G , the first term (Borel pole) suppresses the explosion, while the second term (trans-series instanton) injects a non-perturbative jump that guides the optimization toward better basins. For small G , $\alpha(G) \approx 1$, leaving AdamW untouched.

2.4 Transformer Backbone

The backbone consists of 4 stacked blocks, each containing:

- **Geometric Attention:** Multi-head attention with accelerated RoPE, XOR attention bias, Topological MERA score, and a distance-dependent bias constructed from Gaussian and Bessel radial basis functions.
- **Local Convolutional Mixing:** A depthwise 1D convolution followed by a point-wise projection.
- **SwiGLU Feed-Forward:** A gated linear unit with SiLU activation.

The final token representations are fused via an XOR-gated combination of max-pooling and average pooling, and passed to a regression head with 5-fold dropout sampling at inference.

3 Experimental Setup

3.1 Dataset and Preprocessing

We use the QM9 dataset [3] (133,885 organic molecules with up to 9 heavy atoms) in the standard split: 110,000 training, 10,000 validation, and the remainder as test. Each atom is described by 27 fixed physical features (electronegativity, ionization energy, etc.), 20 molecule-dependent features (interatomic distances, local electron density, etc.), and a 16-dimensional atomic-number embedding, totalling 63 dimensions.

The target is the HOMO–LUMO gap (index 4), normalized to zero mean and unit variance on the training set. Metrics are reported in the original eV scale.

3.2 Training Details

All phases use the AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4} , and a cosine learning-rate schedule with 5% linear warmup. The fine-tuning phase runs for 60 epochs with batch size 128 and a maximum learning rate of 10^{-3} , decreasing to 5×10^{-5} . Exponential Moving Average ($\beta = 0.999$) is applied to the model weights, and Adversarial Weight Perturbation [4] is activated from epoch 15. The Lookahead wrapper was removed to accelerate convergence.

4 Results

Table 1 summarizes the final performance on the test set. The model selected by best validation MAE (0.0573 eV) yields even stronger test metrics. A scatter plot of predicted vs. true values (available in the project repository) confirms near-perfect correlation across the full energy range.

Table 1: Test set results for HOMO–LUMO gap regression on QM9.

Metric	Value
MAE (eV)	0.057553
RMSE (eV)	0.082392
R^2	0.995945
Pearson r	0.998006
Median AE (eV)	0.041225
Max AE (eV)	0.933105
Best Val. MAE (eV)	0.057267
Best Epoch	60
Parameters	2,841,737 (2.84M)

5 Reproducibility

The complete source code is available at:

<https://github.com/benficaized306-web/GrafoPropagation> (file QM9.py).

To reproduce the results:

1. Open a Kaggle notebook (or any GPU-enabled environment).
2. Install dependencies:

```
pip install torch_geometric rdkit
```

3. Copy the contents of QM9.py into a cell and run.

The script automatically downloads QM9, performs all preprocessing, runs the three training phases, and outputs the final metrics.

Pre-trained checkpoints are also provided in the repository:

- `prepretrained_backbone.pt`
- `pretrained_backbone.pt`
- `best_model.pt`

6 Conclusion

We introduced GrafoPropagation v29, a geometric transformer that achieves a test MAE of 0.0576 eV on the QM9 HOMO–LUMO gap without relying on explicit equivariance constraints. The model’s strong performance arises from three key components: asymmetric BYOL pre-training, a Spherical Shock Terrain for position encoding, and a Borel–Laplace Resurgent Optimizer. With only 2.84 million parameters and a fully public codebase, this work demonstrates that human-conceived logic, refined through AI collaboration, can produce cutting-edge scientific results.

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References

- [1] C. M. S. Fernandes, *GrafoPropagation NLP: A Graph-Based Reasoning Architecture*, Zenodo, 2024. <https://zenodo.org/records/20446506>
- [2] J. B. Grill *et al.*, *Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning*, NeurIPS, 2020.
- [3] R. Ramakrishnan *et al.*, *Quantum chemistry structures and properties of 134 kilo molecules*, Scientific Data, 2014.
- [4] D. Wu, S. T. Xia, Y. Wang, *Adversarial Weight Perturbation Helps Robust Generalization*, NeurIPS, 2020.